Market Basket Analysis

Optimizing Retail Sales



Data Mining and Business Intelligence (DMBI)
Continuous Assessment 2 · Mini Project · Class D15A

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Problem → Data → Preprocessing → Algorithm → Results → Evaluation → Insights → Decision

The Business Problem

THE STORY

A supermarket sells thousands of products but does not know which ones customers buy together, so shelves and offers are guesswork.

Objective: find products bought together, then create bundle offers and shelf layouts that lift revenue per trip.

01

Find hidden buying patterns

02

Turn patterns into rules

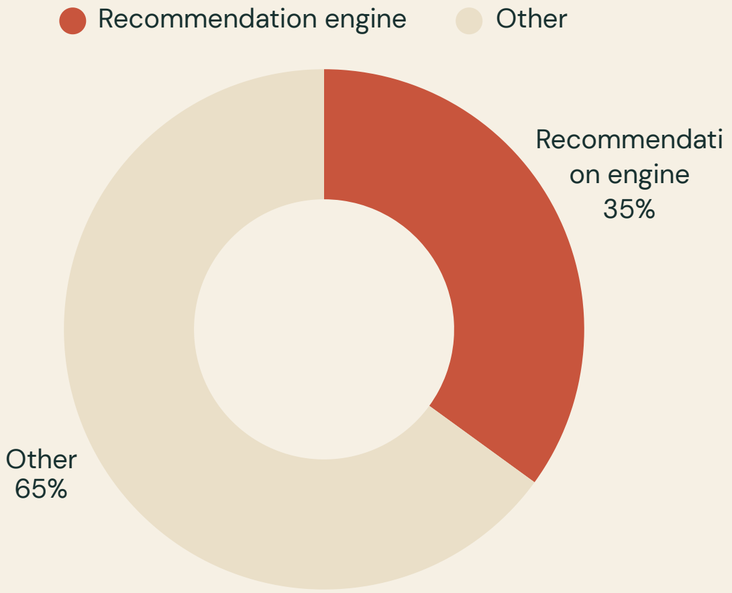
03

Turn rules into store decisions

PROBLEM › DATA › PREPROCESSING › ALGORITHM › RESULTS › EVALUATION › INSIGHTS › DECISION

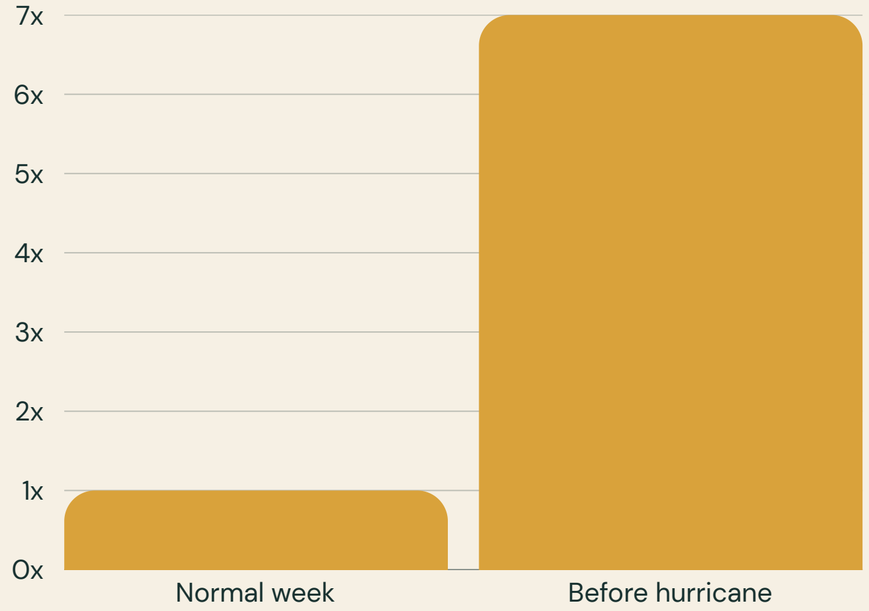
Task: Association Rule Mining · unsupervised

Why It Matters: Real-World Proof



35%

of Amazon sales attributed to its recommendation engine (McKinsey estimate, widely reported)



7×

Walmart 2004: strawberry Pop-Tarts sold at about seven times normal rate before Hurricane Frances; beer was the top pre-hurricane item.

Same method. Same value. At any scale.

Sources: McKinsey estimate, widely reported · New York Times, 14 Nov 2004, Linda M. Dillman

Theory: Association Rule Mining

A rule A → B means baskets with A often contain B. It shows association, not causation.

SUPPORT = count(A and B) / N

how popular

CONFIDENCE = Support(A,B) / Support(A)

how reliable

LIFT = Confidence / Support(B)

how strong

< 1 negative

= 1 independent

> 1 positive

EXAMPLE

Bread 20% × Milk 30% predicts 6% together. Observed 12% → lift = 2.

Theory: Apriori and FP-Growth

APRIORI · Agrawal and Srikant, VLDB 1994

Principle: every subset of a frequent itemset must be frequent, so candidates with an infrequent subset are pruned.

Steps: count singles → keep frequent → join pairs → prune → repeat

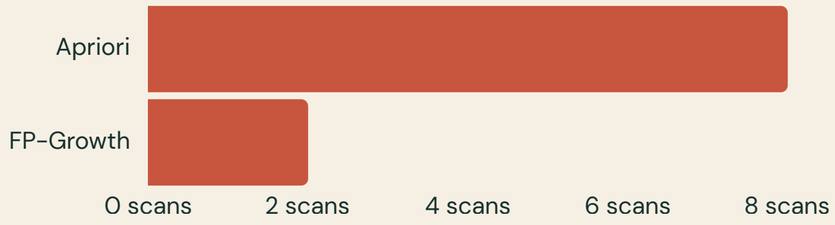
Cost: many database passes

3× to 10×+ faster than earlier algorithms (original paper)

FP-GROWTH · Han et al., 2000

Compress baskets into an FP-tree, mine it recursively, no candidate generation, only two database passes.

Faster on large data; tree needs memory.

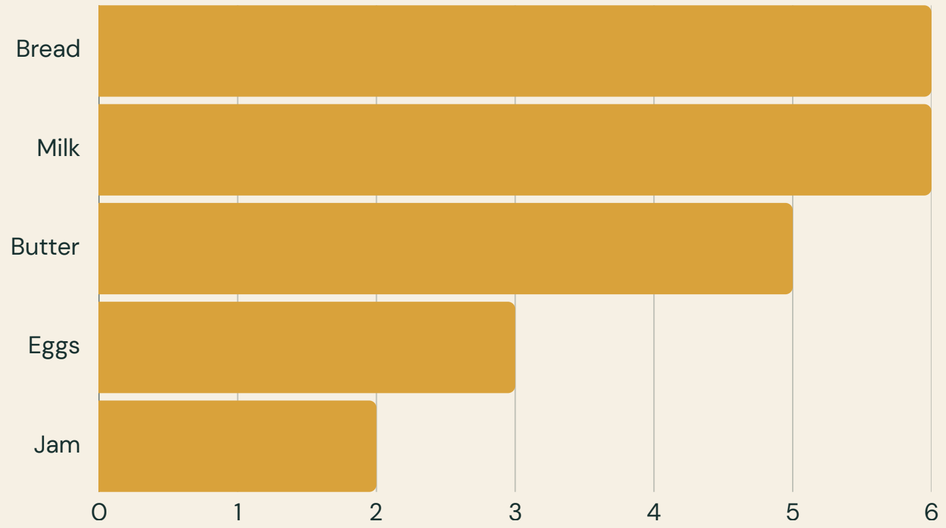


Both return exactly the same frequent itemsets.
Only speed and memory differ.

Worked Example: 8 Baskets by Hand

TRANSACTIONS

BASKET	ITEMS
1	Bread · Butter · Milk
2	Bread · Butter
3	Bread · Milk · Eggs
4	Bread · Butter · Milk · Jam
5	Milk · Eggs
6	Bread · Butter · Jam
7	Bread · Milk
8	Butter · Milk · Eggs



MIN SUPPORT 3/8 = 37.5%

Step 2 pairs kept: Bread+Milk 4, Bread+Butter 4, Butter+Milk 3, Eggs+Milk 3. Pruned: Bread+Eggs 1, Butter+Eggs 1.
Step 3: Bread+Butter+Milk = 2 dropped; search ends.

RULES · min confidence 60%

Eggs → Milk
37.5% support · 100% confidence · lift 1.33

Butter → Bread
80% confidence · lift 1.07

Bread → Milk
66.7% confidence · lift 0.89

TAKEAWAY

Bread and Milk are popular, yet lift 0.89 shows no real link.

Dataset and Preprocessing

9,835
BASKETS

169
PRODUCTS

2.6%
MATRIX DENSITY

<5
ITEMS / BASKET

01 LOAD
one basket per row

02 CLEAN
trim · lowercase

03 ENCODE
TransactionEncoder

04 OUTPUT
9,835 × 169 Boolean

SAMPLE BOOLEAN MATRIX

BASKET	MILK	BREAD	BUTTER	EGGS
1	TRUE	TRUE	TRUE	FALSE
2	FALSE	TRUE	TRUE	FALSE
3	TRUE	TRUE	FALSE	TRUE
4	TRUE	FALSE	TRUE	TRUE

WHY A BOOLEAN MATRIX?

mlxtend needs one column per product.

Groceries dataset, one month of real transactions; Hahsler et al., arules; also groceries.csv on Kaggle.

Implementation in Python

```
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, fpgrowth, association_rules

te = TransactionEncoder()
df = pd.DataFrame(te.fit(baskets).transform(baskets), columns=te.columns_)
freq = apriori(df, min_support=0.01, use_colnames=True)
freq_fp = fpgrowth(df, min_support=0.01, use_colnames=True)
rules = association_rules(freq_fp, metric='confidence', min_threshold=0.25)
rules.sort_values('lift', ascending=False)
```

PARAMETERS

min support 1% = at least 98 baskets;
min confidence 25%

WHY THESE THRESHOLDS

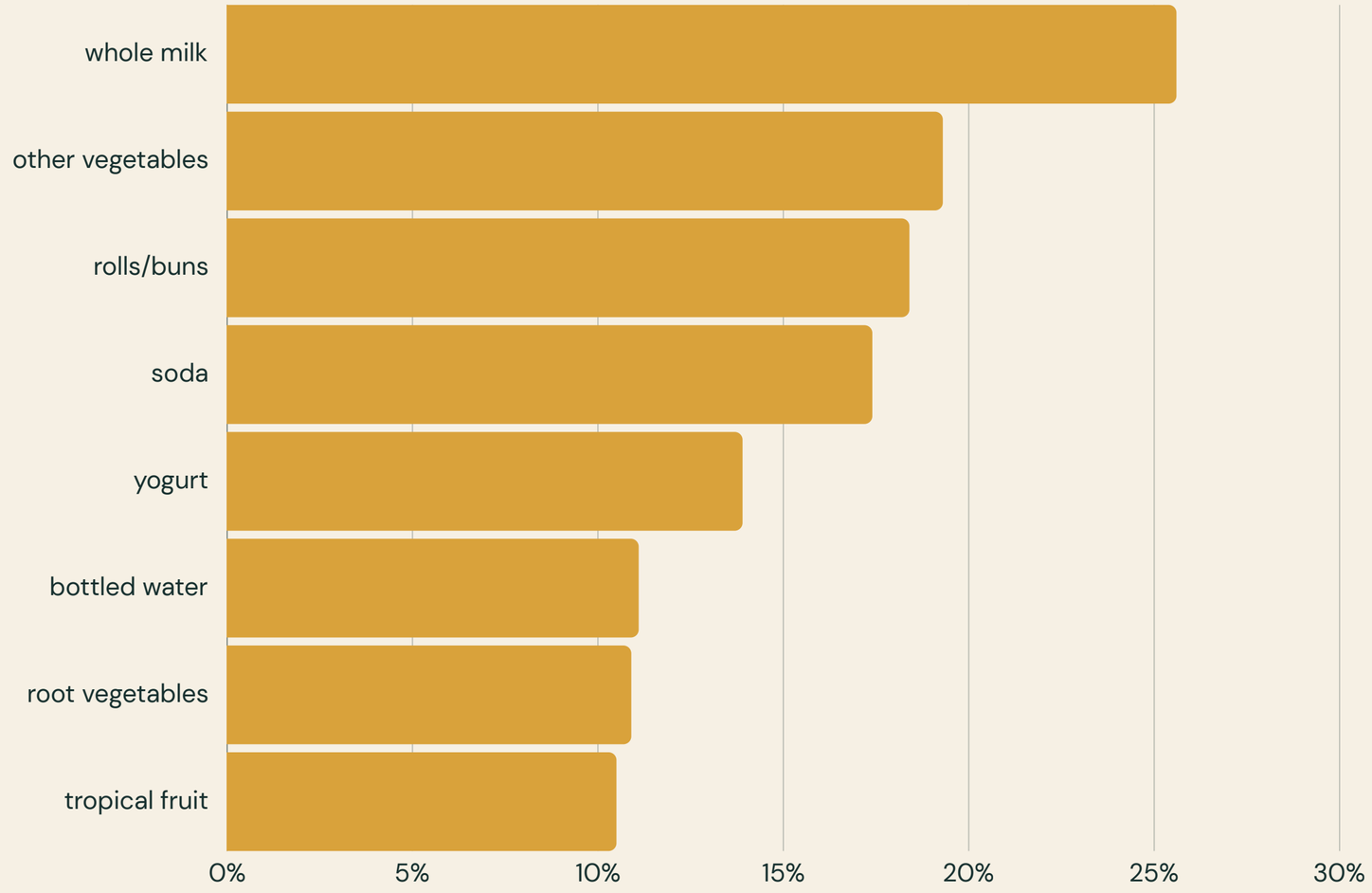
Low support keeps rare but useful pairs; confidence filters weak rules.

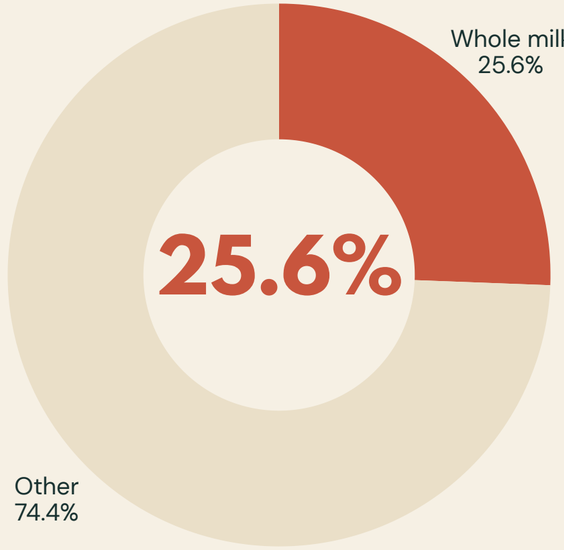
RESULT · 170 RULES

Apriori and FP-Growth return the same itemsets.

Reference run on Groceries, supp 0.01, conf 0.25 · books.psychstat.org Market Basket Analysis

What Do Shoppers Buy Most





HOW TO READ

Whole milk is in 2,513 of 9,835 baskets

SO WHAT

Popular is not the same as linked — which is why we need lift.

Top 10 Rules Ranked by Lift

	RULE	SUPPORT	CONFIDENCE	LIFT
1	**citrus fruit + other vegetables → root vegetables**	**1.04%**	**35.9%**	**3.30**
2	**other vegetables + tropical fruit → root vegetables**	**1.23%**	**34.3%**	**3.14**
3	**beef → root vegetables**	**1.74%**	**33.1%**	**3.04**
4	citrus fruit + root vegetables → other vegetables	1.04%	58.6%	3.03
5	root vegetables + tropical fruit → other vegetables	1.23%	58.5%	3.02
6	other vegetables + whole milk → root vegetables	2.32%	31.0%	2.84
7	curd + whole milk → yogurt	1.01%	38.5%	2.76
8	other vegetables + yogurt → root vegetables	1.29%	29.7%	2.73
9	other vegetables + yogurt → tropical fruit	1.23%	28.3%	2.70
10	other vegetables + rolls/buns → root vegetables	1.22%	28.6%	2.63

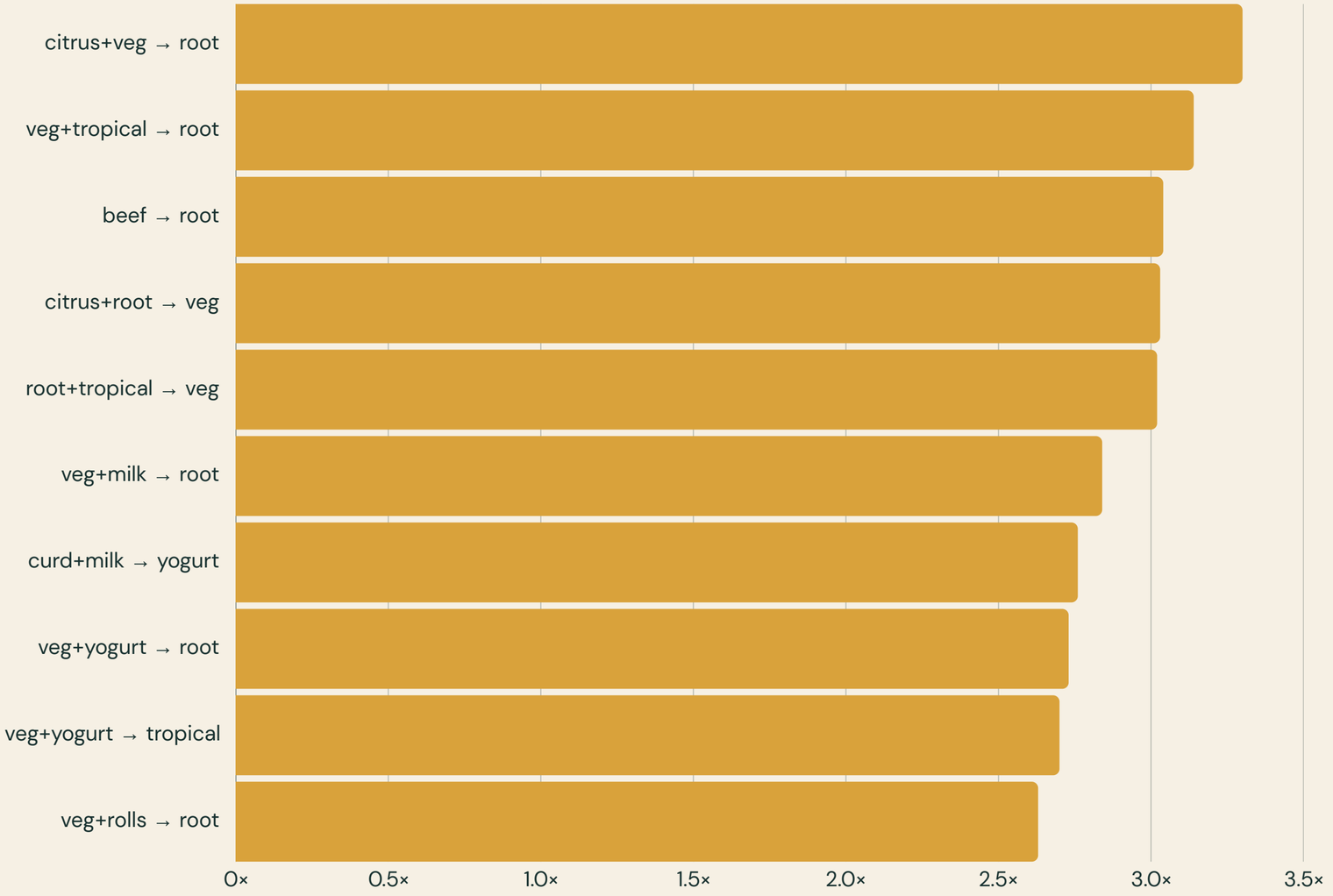
PATTERN

Root vegetables is the consequent in 6 of the top 10 rules and appears in 8 of 10.

READ ROW 1

35.9% of baskets with citrus fruit and other vegetables also hold root vegetables. Lift 3.30 = 3.3 times more likely than average.

Lift: How Strong Is the Link



3.30×

Shoppers who buy citrus fruit and other vegetables are 3.3 times more likely to add root vegetables than an average shopper.

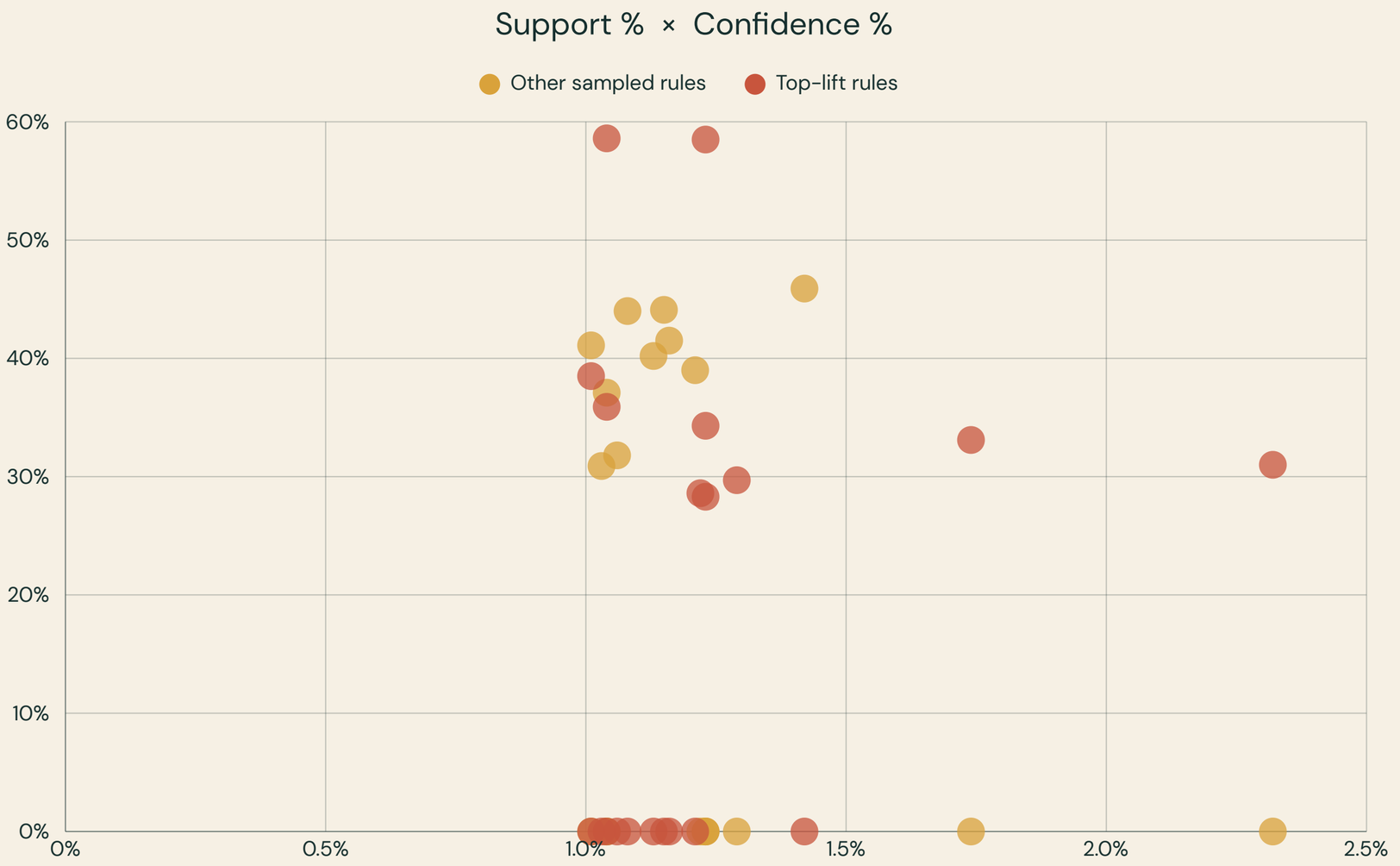
BASELINE

Lift = 1 means independence

WHY LIFT BEATS CONFIDENCE

Confidence is inflated by popular items; lift corrects for that.

Support vs Confidence



20 of the 170 rules, published Groceries reference run.

READING GUIDE

Right = more common; top = more reliable.

SWEET SPOT

Top-right: frequent and reliable; high lift adds evidence of real affinity.

TRADE-OFF

Rules near 58% confidence sit near 1% support: reliable but few baskets.

Product Network Graph

HUB

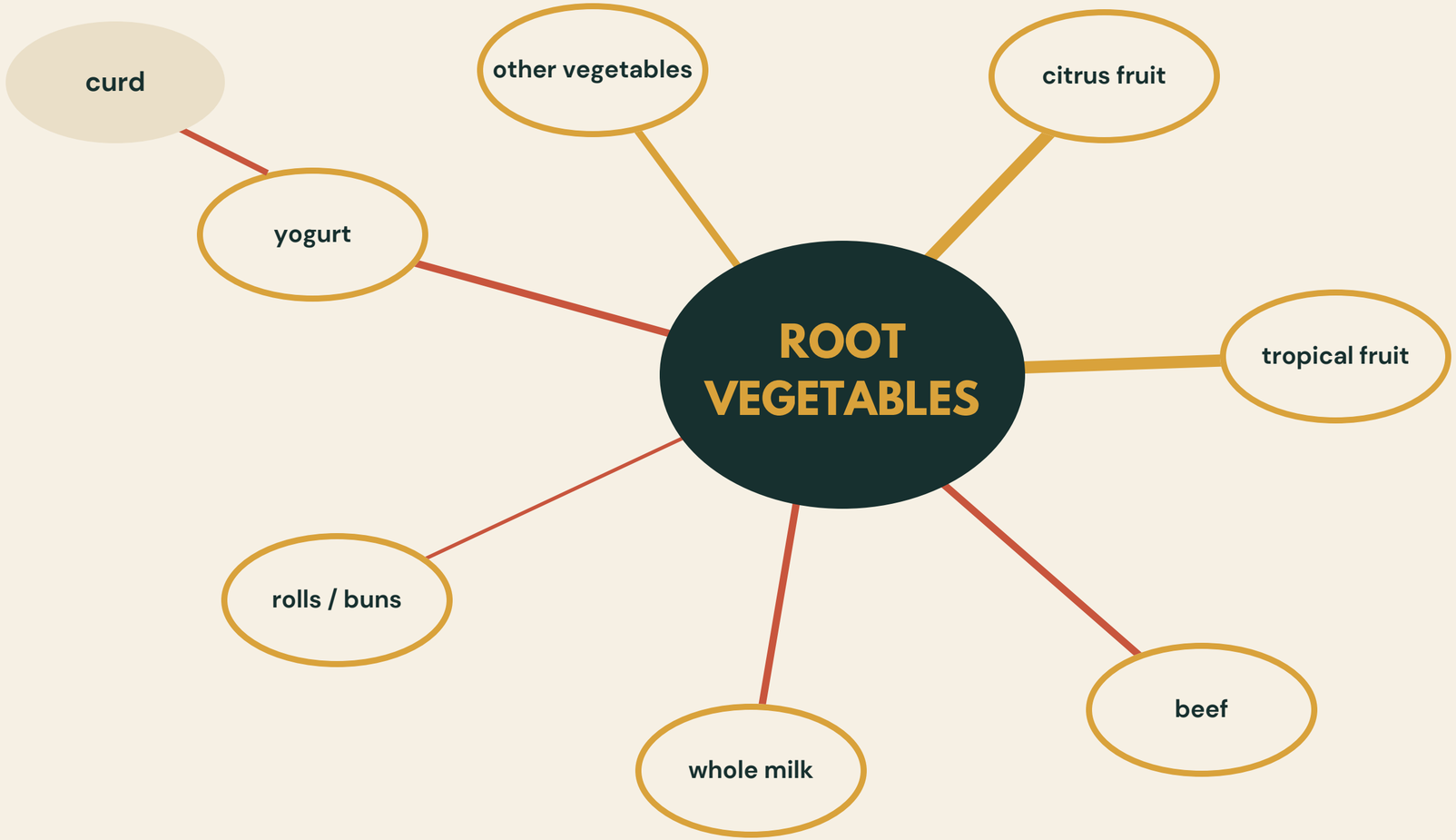
Root vegetables connects most products

CLUSTER 1

Fresh produce forms the strongest community

CLUSTER 2

Dairy links curd, whole milk and yogurt



The network shows structure that a table hides.

Insights and Store Decisions

INSIGHTS

01

Fresh produce is the strongest cluster: root vegetables, other vegetables, citrus and tropical fruit pull each other in.

02

Popular is not linked: whole milk is in 25.6% of baskets, yet sample rules ending in whole milk have lift only about 1.5–1.7.

03

Dairy cross-link: curd + whole milk predicts yogurt, lift 2.76.

DECISIONS

01

SHELF LAYOUT — one produce zone with root vegetables beside citrus, tropical fruit and other vegetables

02

BUNDLE — ‘Fresh Veg Bundle’ at a small discount

03

CHECKOUT COUPON — yogurt coupon for curd + whole milk shoppers

A/B TEST → average basket value • items per basket • bundle attach rate
Impact is measured, not assumed.

Association rule mining on 9,835 baskets produced 170 rules; lift ranked them into clear store actions.

LIMITATIONS

Depends on thresholds; correlation not causation; one store and one month; no price or margin data.

FUTURE SCOPE

Add margins, time of day, customer segments and sequence mining.

Questions?

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6 Raschka S, mlxtend documentation.
7 Han J et al., Mining Frequent Patterns without Candidate Generation, SIGMOD 2000.